

ReflectionPad3d#

[https://pytorch.org/docs/stable/generated/torch.nn.ReflectionPad3d.html#torch](https://pytorch.org/docs/stable/generated/torch.nn.ReflectionPad3d.html#torch.nn.ReflectionPad3d)

Description:

Pads the input tensor using the reflection of the input boundary. For N-dimensional padding, use `torch.nn.functional.pad()`. padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 6-tuple, uses (`padding_left\text{padding\_left}`padding_left, `padding_right\text{padding\_right}`padding_right, `padding_top\text{padding\_top}`padding_top, `padding_bottom\text{padding\_bottom}`padding_bottom, `padding_front\text{padding\_front}`padding_front, `padding_back\text{padding\_back}`padding_back) Note that padding size should be less than the corresponding input dimension. Input: $(N, C, D_{in}, H_{in}, W_{in})$ or $(N, C, D_{in}, H_{in}, W_{in})$ or $(C, D_{in}, H_{in}, W_{in})$. Output: $(N, C, D_{out}, H_{out}, W_{out})$ or $(N, C, D_{out}, H_{out}, W_{out})$ or $(C, D_{out}, H_{out}, W_{out})$, where

Function Signature

```
class torch.nn.ReflectionPad3d(padding)[source]#
```

Parameters

Shape:

Code Examples

```
>>> m = nn.ReflectionPad3d(1)
>>> input = torch.arange(8, dtype=torch.float).reshape(1, 1, 2,
2, 2)
>>> m(input)
tensor([[[[ [7., 6., 7., 6.],
            [5., 4., 5., 4.],
            [7., 6., 7., 6.],
            [5., 4., 5., 4.]],
          [[3., 2., 3., 2.],
            [1., 0., 1., 0.],
            [3., 2., 3., 2.],
            [1., 0., 1., 0.]],
          [[7., 6., 7., 6.],
            [5., 4., 5., 4.],
            [7., 6., 7., 6.],
            [5., 4., 5., 4.]],
          [[3., 2., 3., 2.],
            [1., 0., 1., 0.],
            [3., 2., 3., 2.],
            [1., 0., 1., 0.]]]])])
```

Detailed Documentation

Documentation

Pads the input tensor using the reflection of the input boundary.

For N-dimensional padding, use `torch.nn.functional.pad()`.

padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 6-tuple, uses (`padding_left` $\text{\texttt{\text{padding_left}}}$ `padding_left`, `padding_right` $\text{\texttt{\text{padding_right}}}$ `padding_right`, `padding_top` $\text{\texttt{\text{padding_top}}}$ `padding_top`, `padding_bottom` $\text{\texttt{\text{padding_bottom}}}$ `padding_bottom`, `padding_front` $\text{\texttt{\text{padding_front}}}$ `padding_front`, `padding_back` $\text{\texttt{\text{padding_back}}}$ `padding_back`) Note that padding size should be less than the corresponding input dimension.

Input: (N,C,D_{in},H_{in},W_{in})(N, C, D_{in}, H_{in}, W_{in})(N,C,D_{in},H_{in},W_{in}) or (C,D_{in},H_{in},W_{in})(C, D_{in}, H_{in}, W_{in})(C,D_{in},H_{in},W_{in}).

Output: (N,C,D_{out},H_{out},W_{out})(N, C, D_{out}, H_{out}, W_{out})(N,C,D_{out},H_{out},W_{out}) or (C,D_{out},H_{out},W_{out})(C, D_{out}, H_{out}, W_{out})(C,D_{out},H_{out},W_{out}), where

$D_{out} = D_{in} + \text{\texttt{\text{padding_front}}} + \text{\texttt{\text{padding_back}}}$
 $D_{out} = D_{in} + \text{\texttt{\text{padding_front}}} + \text{\texttt{\text{padding_back}}}$

$H_{out} = H_{in} + \text{\texttt{\text{padding_top}}} + \text{\texttt{\text{padding_bottom}}}$
 $H_{out} = H_{in} + \text{\texttt{\text{padding_top}}} + \text{\texttt{\text{padding_bottom}}}$

$W_{out} = W_{in} + \text{\texttt{\text{padding_left}}} + \text{\texttt{\text{padding_right}}}$
 $W_{out} = W_{in} + \text{\texttt{\text{padding_left}}} + \text{\texttt{\text{padding_right}}}$

`\text{padding\_right}Wout=Win+padding\_left+padding\_right`